

HOMEWORK 10

6500-324: DATA MANAGEMENT FOR INFORMATION SYSTEMS (Fall 2022)

©2022 by ecourse.org

Part 1. Multiple Choice or True/False

- (1 point) Delete the employee who has the longest employment in the company
A) delete from emp where (hiredate, deptno) in (select max(hiredate), deptno from emp group by deptno);
B) delete from emp where hiredate in (select max(hiredate) from emp);
C) delete from emp where hiredate in (select min(hiredate) from emp);
D) delete from emp s where s.hiredate in (select min(t.hiredate) from emp t where t.deptno = s.deptno);
E) delete from emp s where s.hiredate = (select min(t.hiredate) from emp t);
- (1 point) For all the employees supervised by KING, give them a salary increase, which is equal to the 2% of the company's average salary

Code A:

```
<pre>
update emp
set sal =
  (select 1.02 * avg(sal) from emp)
where mgr in
  (select empno from emp where ename = 'KING');
</pre>
```

Code B:

```
<pre>
update emp
set sal =
  (select avg(1.02*sal) from emp)
where mgr in
  (select empno from emp);
</pre>
```

Code C:

```
<pre>
update emp a, emp b
set a.sal =
  (select avg(1.02*a.sal) from emp)
where a.mgr in
  (select empno from emp b where b.ename = 'KING');
</pre>
```

- A) A, B, and C
B) A and B
C) B and C
D) A and C
E) A only

3. (1 point) Function decode(comm, null, 0, comm) is the same as
A) NVL(comm, 0)
B) decode(comm, comm, comm, 0)
C) update emp set comm = 0 where comm is null;
D) update comm = 0 when comm is null
4. (1 point) Base on the tables shown in the figure below, give 2% salary increase to those who have the longest employment in their corresponding department.

Code A.

```
<pre>
update emp a
set a.sal = 1.02 * a.sal
where a.hiredate = (select min(b.hiredate)
from emp b
where b.deptno = a.deptno);
</pre>
```

Code B:

```
<pre>
update emp
set sal = 1.02 * sal
where (deptno, hiredate) in
(select deptno, min(hiredate)
from emp
group by deptno);
</pre>
```

Code c:

```
<pre>
update emp
set sal = 1.02 * sal
where empno in
(select empno from emp a
where a.hiredate =
(select min(b.hiredate) from emp b
where b.deptno = a.deptno));
</pre>
```

A) A, B, and C

- B) A and B
C) B and C
D) A and C
E) C only

5. (1 point) Find one department at random based on the following tables:
A) select * from dept order by dbms_random.value;
B) select * from (select * from dept order by dbms_random.value) where rownum < 2;
C) select deptno from emp where rownum < 2;
D) select deptno from (select deptno from emp) where rownum < 2;

6. (1 point) (select all that apply) Find jobs whose minimum salary is less than 2000 and maximum salary is over 4000
- A) (select job from emp group by job having min(sal) < 2000) union (select job from emp group by job having max(sal) > 4000);
- B) select job from emp group by job having min(sal) < 2000 and max(sal) > 4000;
- C) (select job from emp group by job having min(sal) < 2000) minus (select job from emp group by job having max(sal) > 4000);
- D) (select job from emp group by job having min(sal) < 2000) intersect (select job from emp group by job having max(sal) > 4000);
- E) select job from emp group by job having min(sal) < 2000 and having max(sal) > 4000;

7. (1 point) What is wrong with the following to find the total salary?

```
<pre>
select sum(sal) from (
(select sal from emp where comm is null)
union
(select sal from emp where comm is not null)
)
</pre>
```

- </pre>
- A) there is no need to use union
- B) after from must be a table, not a subquery
- C) the union will remove duplicate salaries
- D) nothing wrong

8. (1 point) (select all that apply) Find the department located in Dallas and also have over two employees.
- A) select dname from dept, emp where dept.deptno = emp.deptno and loc='DALLAS' group by dname having count(*) > 2;
- B) select dname from dept, emp where dept.deptno = emp.deptno and loc='DALLAS' group by deptno having count(*) > 2;
- C) (select deptno from emp group by deptno having count(*) > 2) intersect (select deptno from dept where loc='DALLAS');
- D) (select * from emp group by deptno having count(*) > 2) intersect (select * from dept where loc='DALLAS');
9. (1 point) Based on the following tables, what is the correct statement to find employees who makes less than the average in there corresponding job titles?
- A) select ename from em where (job, sal) < (select job, avg(sal) from emp group by job);
- B) select a.ename from emp a where a.sal < (select avg(b.sal) from emp b where b.deptno = a.deptno);
- C) select ename from emp a where a.sal < (select avg(b.sal) from emp b where b.job = a.job);
- D) select ename from em where (deptno sal) < (select deptno, avg(sal) from emp group by deptno);
10. (1 point) Which of the following is the correct criteria to search for a word in a text column based on the sound of the word?

- A) contains(myTextColumn, '?support') > 0
- B) contains(myTextColumn, '\$support') > 0
- C) contains(myTextColumn, '!support') > 0
- D) contains(myTextColumn, 'support') > 0